Karthik Uppari

Software Development Intern

Hyderabad, India | upparikarthik505@gmail.com | linkedin.com/in/karthik-uppari-4005a7373 | github.com/karthik1122-code

PROFESSIONAL SUMMARY

B.Tech Computer Science (AI/ML) student at NIAT with hands-on experience building full-stack web applications using React, Node.js, Express.js and MongoDB, including REST APIs, JWT authentication and retrieval-augmented generation (RAG) systems. Interested in full-stack and backend development. Seeking an SDE / Full Stack internship to apply these skills to real-world projects.

TECHNICAL SKILLS

Languages:	JavaScript, Python, C++
Frontend:	React, HTML, CSS
Backend:	Node.js, Express.js, REST APIs, JWT Authentication
Database:	MongoDB, Mongoose, PostgreSQL
Generative AI:	Prompt Engineering, LLM APIs (Gemini), RAG, Output Validation
Tools:	Git, GitHub, Vercel, Render

PROJECTS

SpendLog - Expense tracker with user accounts

React | Node.js | Express.js | MongoDB | Mongoose | JWT

- Built a full-stack expense tracker with a React interface, category filter, monthly total and spending breakdown, backed by a REST API on Express.js and MongoDB (Mongoose).
- Implemented sign-up and login with bcrypt-hashed passwords and JWT stored in httpOnly cookies, with protected routes.
- Scoped every query to the logged-in user so one user cannot read or delete another user's data; covered by automated tests.
- Deployed live on Render with MongoDB Atlas.

GitHub: github.com/karthik1122-code/spendlog | Live Demo: spendlog-e525.onrender.com

DocuMind - Evaluation-first RAG over the Express docs

Node.js | BM25 | LSA | Reciprocal Rank Fusion

- Wrote a 50-question test set before tuning; heading-based chunking beat fixed windows (recall@1 65.7% vs 51.4%).
- Combined BM25 and LSA with reciprocal rank fusion and a reranker for retrieval.
- Answers cite their source passage and stream over SSE; refused all 15 unanswerable questions; 29 automated tests.

GitHub: github.com/karthik1122-code/documind

AgentGate - Policy gate for AI agent tool calls

Node.js | PostgreSQL | Policy engine | Hash-chained audit log

- Built a default-deny gate for AI agent tool calls: each call is checked against a policy before it can run.
- Tied approvals to the exact call, required a different approver, and made approvals expire.
- Recorded every decision in a hash-chained audit log in PostgreSQL; passwords hashed with scrypt, with role-based access.
- Tested with a 56-case adversarial suite (author-written, simulated tools): 0 unauthorised actions executed.

GitHub: github.com/karthik1122-code/agentgate | Live Demo: agentgate-lake.vercel.app

EXPERIENCE

IoT & Robotics Trainee - T-Works — Aug - Sep 2024

- Worked on AI-powered smart street lights: hardware integration, sensor calibration and IoT connectivity.

Student Intern - Airbac Labs — Apr - Jul 2024

- Learned Python and machine learning and applied them to practical projects, including stock-market prediction models.

Community Volunteer - Rubaroo — Aug 2024

- Led a STEM-career awareness project that engaged 100+ students.

EDUCATION

B.Tech, Computer Science (AI/ML) - NxtWave Institute of Advanced Technologies (NIAT), Hyderabad — 2025 - 2029

SGPA: 9.52 (Semester 3)

Intermediate (Class XII) - Texas International — 2023 - 2025

Percentage: 90.5%